

# Physics-Informed ICU Bed Forecasting with Cost-Asymmetric Quantile Loss and Robust Optimisation

For NHS Critical-Care Surge Capacity Under Demand Uncertainty

**Michael Ajao-Olarinoye** · Abiola Babatunde · Vasile Palade

Centre for Computational Sciences and Mathematical Modelling, Coventry University

UKCI 2026 · Coventry · 9–11 September 2026

# The Problem: Planning Surge Capacity Before Demand Is Known

---

During an epidemic wave, NHS planners must **pre-position critical-care beds** and **route patients** across the **seven NHS England regions** — committing *before* demand is observed.

- **Regime shift:** Delta → Omicron changed the case-to-hospitalisation ratio, breaking stationary models
- **Operating limits:** finite surge budget ( $\approx 20\%$  of baseline) and travel-time-capped mutual aid ( $\leq 240$  min)

**The error cost is asymmetric.** Under-prediction strands patients (high cost); over-prediction only wastes prepared capacity (low cost). Standard forecasters optimise *symmetric* accuracy.

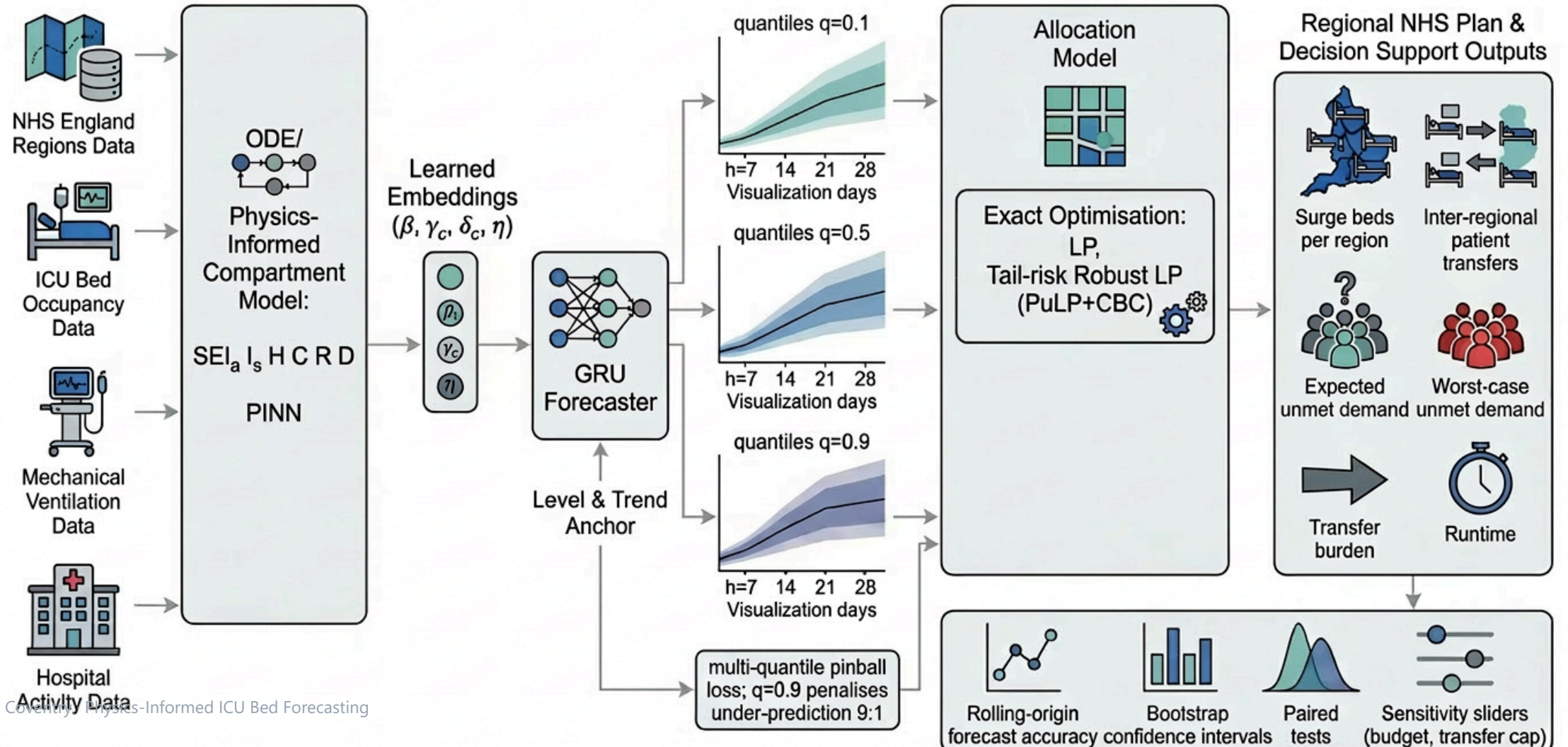
# Contributions

---

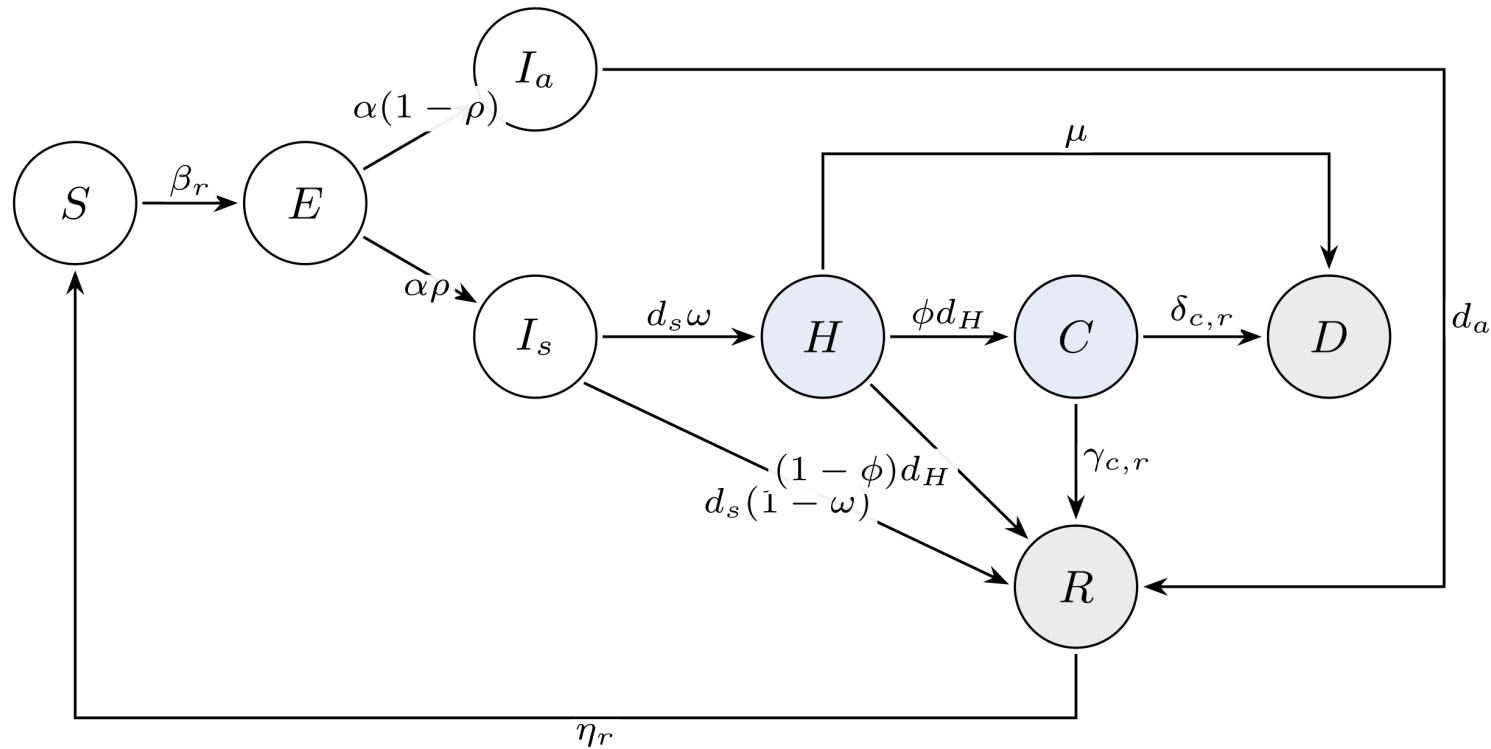
1. **Cost-asymmetric forecaster** — a multi-quantile pinball loss (the  $q^{0.9}$  branch penalises under-prediction **9-to-1**) plus a learnable level-and-trend anchor. Cuts 14-day RMSE **19%** vs refitted ARIMA, **37%** vs a per-region GRU.
2. **Robust LP allocation** — a hybrid-regional tail-risk linear program that attains the **lowest transfer burden** ( $4.6\times$  below population-proportional) among budget-exhausting policies.
3.  **$SEI_a I_s HCRD$  clinical split** — separates hospitalisation  $\omega$  ( $I_s \rightarrow H$ ) from critical-care escalation  $\phi$  ( $H \rightarrow C$ ), fixing the overloaded  $\omega$  of prior work.

**Gap closed:** no prior work combines physics-informed *regional* NHS ICU forecasting with cost-asymmetric quantile training and an *exact robust LP* across all seven regions.

# A Forecast-to-Decision Pipeline



# The $SEI_a I_s HCRD$ Compartmental Model



- Eight compartments; only  $H$  (hospitalised) and  $C$  (critical-care) are observed at regional resolution (shaded)
- **Key fix:** distinct  $\omega$  ( $I_s \rightarrow H$ ) and  $\phi$  ( $H \rightarrow C$ ) make the two clinical thresholds independently identifiable

## Local Epidemic Dynamics (per region $r$ )

---

$$\begin{aligned}\dot{S}_r &= -\beta_r S_r (I_{s,r} + I_{a,r}) / N_r + \eta_r R_r, & \dot{E}_r &= \beta_r S_r (I_{s,r} + I_{a,r}) / N_r - \alpha E_r, \\ \dot{I}_{s,r} &= \alpha \rho E_r - d_s I_{s,r}, & \dot{I}_{a,r} &= \alpha (1 - \rho) E_r - d_a I_{a,r}, \\ \dot{H}_r &= d_s \omega I_{s,r} - (d_H + \mu) H_r, & \dot{C}_r &= \phi d_H H_r - (\gamma_{c,r} + \delta_{c,r}) C_r, \\ \dot{R}_r &= d_s (1 - \omega) I_{s,r} + d_a I_{a,r} + (1 - \phi) d_H H_r + \gamma_{c,r} C_r - \eta_r R_r, \\ \dot{D}_r &= \mu H_r + \delta_{c,r} C_r,\end{aligned}$$

- Fixed clinical constants  $\alpha, d_s, d_a, d_H, \rho, \omega, \mu, \phi$ ; **learned per-region**  $\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r$
- Conserves population:  $S_r + E_r + I_{a,r} + I_{s,r} + H_r + C_r + R_r + D_r = N_r$

## Phase A — PINN Pre-Training

---

Two MLPs per region: a **state network**  $U_{NN}^r$  ( $5 \times 20$ , the 8 states) and a **parameter network**  $X_{NN}^r$  ( $3 \times 20$ , the 4 learned rates).

$$\mathcal{L}_r^{PINN} = \underbrace{\sum_t \sum_{k \in \{H, C\}} (U_{NN,k}^r(t) - y_{r,t,k}/N_r)^2}_{\text{data fit on observed } H, C} + \lambda_{ode} \underbrace{\sum_t \left\| \frac{dU_{NN}^r}{dt} - \mathcal{N}(U_{NN}^r; X_{NN}^r) \right\|^2}_{\text{ODE residual (256 collocation pts)}}$$

- 1500 Adam steps,  $\lambda_{ode}=0.1$ ; both terms converge to  $\sim 4 \times 10^{-6}$
- With only  $H, C$  observed the four rates are **not individually identifiable** — consumed as a 4-channel *learned feature*, not interpreted

## Phase A — Temporal Head with Level-and-Trend Anchor

---

The PINN is **frozen**; its 12 outputs + the z-scored target form  $\mathbf{f}_{r,\tau} \in \mathbb{R}^{13}$  over a 28-day window  $\rightarrow$  a **2-layer GRU** (hidden 128)  $\rightarrow$  per-horizon MLP emitting quantile deltas.

$$\hat{z}_{r,t}^{h,q} = \Delta_{r,t}^{h,q} + \alpha^{h,q} \tilde{y}_{r,t} + \gamma^{h,q} s_{r,t}^{(7)} \left(\frac{h}{7}\right)^{\varphi^q}$$

- $\tilde{y}_{r,t}$  = latest observation,  $s_{r,t}^{(7)}$  = one-week slope; **anchor makes the GRU learn a residual**
- Damping exponent  $\varphi^q$  prevents long-horizon slope overshoot ( $\varphi^{0.5} \approx 0.88$ ,  $\varphi^{0.1} = \varphi^{0.9} = 0.5$ )
- Horizons  $h \in \{7, 14, 21, 28\}$ , quantiles  $q \in \{0.1, 0.5, 0.9\}$



## The Cost-Asymmetric Pinball Loss

$$\mathcal{L}_{\text{pinball}} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{E}_{(t,h)} [\max(q e, (q-1) e)], \quad e = \tilde{y}_{r,t+h-1} - \hat{z}_{r,t}^{h,q}$$

- $q=0.5 \rightarrow$  collapses to **L1** (point accuracy)
- $q=0.9 \rightarrow$  penalises **under-prediction 9-to-1** — the operational shortage cost
- $q=0.1 \rightarrow$  the lower interval edge; report  $[\hat{y}^{0.1}, \hat{y}^{0.9}]$  as an 80% band

**Operationally aligned, not solver-coupled.** The loss encodes the shortage asymmetry directly; the forecaster is *not* trained through allocation regret.

Train  $\leq 800$  AdamW epochs on Alpha; early-stop on Delta validation; test on Omicron at 32 rolling origins.

## Phase C — Allocation: Scope, Sets & Parameters

---

**Hybrid-regional:** demand nodes and capacity sites collapse onto the seven NHS regions — every region is "open", cost per surge bed is uniform; the model decides *how many beds where* and *how to route*.

- **Indices:** regions  $r$ ; horizons  $h \in \{7, 14, 21, 28\}$ ; scenarios  $s \in \{\text{low, median, high}\}$ , weights  $\pi = (0.2, 0.6, 0.2)$
- **Capacity:**  $C_r = 1.05 \times \text{Delta-peak}$ ; surge cap  $K_r = 0.5 C_r$ ; **budget**  $B = 0.20 \sum_r C_r = 213 \text{ beds}$
- **Mutual aid:** travel  $T_{rr'}$  from great-circle  $\times 1.3$  at 80 km/h; cap  $\tau = 240 \text{ min} \rightarrow 12$  undirected links (Midlands is the hub)
- **Decisions:** surge  $b_{r,h}$ , peak  $\bar{b}_r$ , transfers  $z_{rr',h}^s$ , unmet  $u_{r,h}^s$ , tail auxiliary  $W$

## Deterministic Median-Scenario LP

---

$$\min \sum_{r,h} u_{r,h}^m + \rho \sum_{r \neq r',h} D_{rr'} z_{rr',h}^m \quad (\rho = 10^{-3}: \text{unmet} \gg \text{routing})$$

$$\sum_{r'} z_{rr',h}^m + u_{r,h}^m \geq d_{r,h}^m \quad (\text{demand: serve, transfer, or record unmet})$$

$$\sum_{r'} z_{r'r,h}^m \leq C_r + b_{r,h} \quad (\text{capacity: baseline + surge})$$

$$b_{r,h} \leq K_r, \quad b_{r,h} \leq \bar{b}_r, \quad \sum_r \bar{b}_r \leq B \quad (\text{peak surge charged once, budget})$$

- Optimises the **median** scenario only; transfers allowed only where  $T_{rr'} \leq \tau$

## Tail-Risk Robust LP + Baselines

---

Impose demand & capacity for **every** scenario; add a tail penalty  $\lambda_3 W$  with  $W \geq \sum_{r,h} u_{r,h}^s$  on the **high** scenario:

$$\min \sum_{s \in \mathcal{S}} \pi_s \left( \sum_{r,h} u_{r,h}^s + \rho \sum_{r \neq r',h} D_{rr'} z_{rr',h}^s \right) + \lambda_3 W$$

- $\lambda_3=0 \rightarrow$  risk-neutral;  $\lambda_3 \rightarrow \infty \rightarrow$  worst-case. **We report  $\lambda_3=1$**
- Solved **exactly** in PuLP+CBC: global optimum in  $< 0.1$  s at 7-region scale
- **Heuristic floors** (re-scored by the same LP oracle): population-proportional, demand-proportional, greedy shortage-first

## Case Study: NHS England, Aug 2020 – Aug 2022

---

**Data (public):** NHS England COVID-19 Hospital Activity (daily regional MV occupancy) · ONS 2021 mid-year populations (PINN normaliser + allocation baseline).

**Chronological split by wave — the held-out wave is the most clinically distinct:**

| Split    | Period              | Wave                      | Role                       |
|----------|---------------------|---------------------------|----------------------------|
| Train    | Aug 2020 – May 2021 | Alpha + early vaccination | fit                        |
| Validate | Jun – Nov 2021      | Delta                     | early stopping / selection |
| Test     | Dec 2021 – Aug 2022 | Omicron                   | regime-shift stress test   |

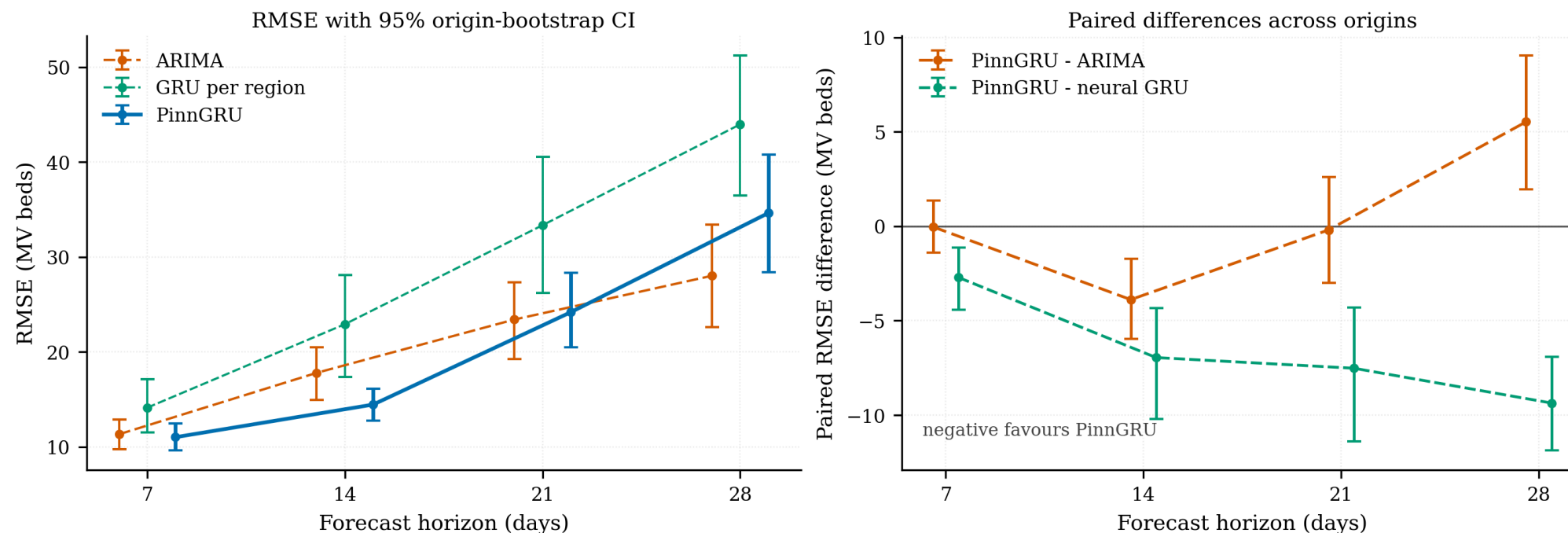
Forecasts macro-averaged over 7 regions × **32 rolling origins** ( $\Delta t = 7$  days); allocation reported at the first Omicron origin (29 Dec 2021).

## Results — Forecast Accuracy

| Model              | h=7   | h=14  | h=21  | h=28  | Under-est. % |
|--------------------|-------|-------|-------|-------|--------------|
| Seasonal-naive(7)  | 13.53 | 18.16 | 22.31 | 25.98 | 37.5         |
| ARIMA per region   | 11.33 | 17.80 | 23.41 | 28.02 | 44.2         |
| XGBoost (lags)     | 12.45 | 22.90 | 36.42 | 58.88 | 27.7         |
| GRU per region     | 14.13 | 22.92 | 33.35 | 43.97 | 7.6          |
| PinnGRU (proposed) | 11.02 | 14.46 | 24.19 | 34.63 | 10.3         |

- Wins **h=7** and **h=14** — the operational surge window. **−19%** vs ARIMA ( $p=0.003$ ), **−37%** vs GRU
- ARIMA's per-origin refit wins **h=28** ( $p=0.006$ ) → PinnGRU is a **short-to-medium-horizon** forecaster (reported honestly, not hidden)

# Results — Forecast Uncertainty



- **Left:** RMSE with 95% origin-bootstrap intervals. **Right:** paired origin-level differences (negative favours PinnGRU)
- PinnGRU significantly better at h=14 (−3.88 beds); ARIMA significantly better at h=28 (+5.53 beds)

## Results — Ablations: What Is Load-Bearing?

| Configuration            | h=14 RMSE | h=28 RMSE | h=14 $\Delta$ |
|--------------------------|-----------|-----------|---------------|
| PinnGRU (full)           | 14.46     | 34.63     | —             |
| w/o cost-asymmetric loss | 18.70     | 41.47     | +29%          |
| w/o PINN pre-training    | 17.27     | 34.73     | +19%          |
| w/o PINN params          | 27.25     | 44.59     | +88%          |

- **PINN parameter features dominate** — zeroing them nearly doubles h=14 RMSE (the largest regression)
- The cost-asymmetric loss improves the RMSE trade-off; PINN pre-training (physics prior, not capacity) drives the gain over a plain GRU

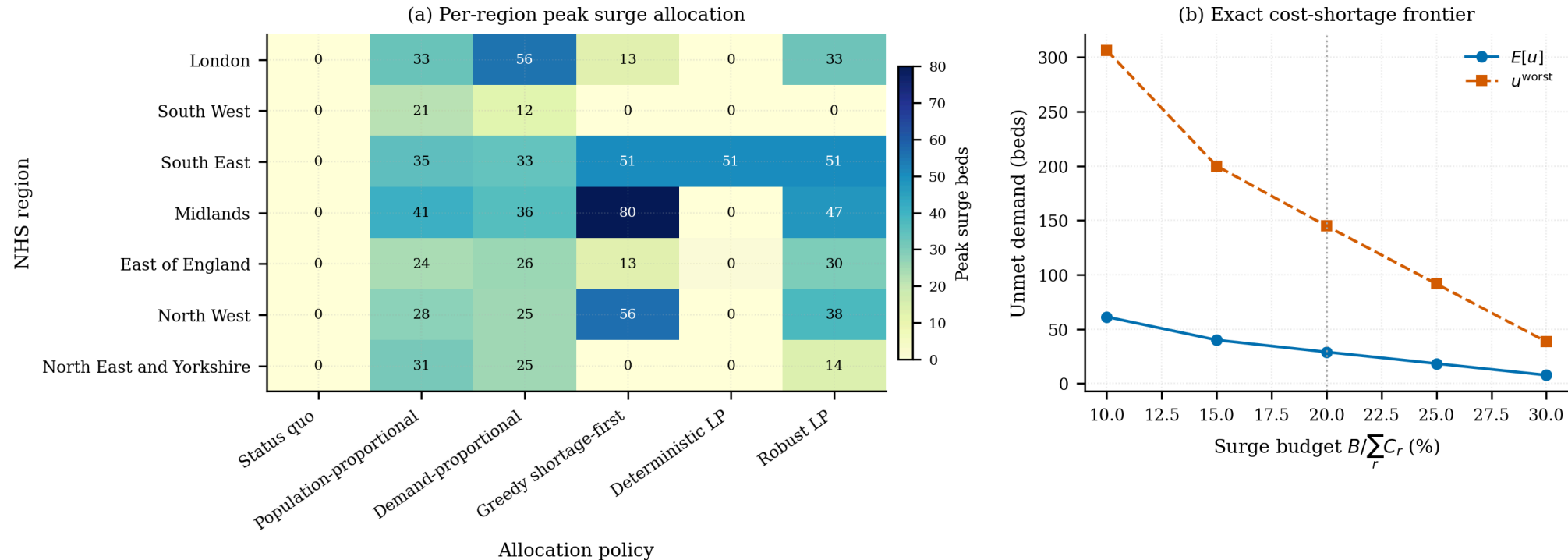


## Results — Allocation (first Omicron origin, B=213)

| Policy                                      | Beds  | $E[u]$ | $u^{\text{worst}}$ | Transfer (bed·km) | Time (s) |
|---------------------------------------------|-------|--------|--------------------|-------------------|----------|
| Population-proportional                     | 212.7 | 29.0   | 144.8              | 3,119.8           | 0.00     |
| Demand-proportional                         | 212.7 | 29.0   | 144.8              | 2,615.3           | 0.00     |
| Greedy shortage-first                       | 212.7 | 29.0   | 144.8              | 1,140.0           | 9.98     |
| Deterministic LP                            | 51.4  | 83.3   | 416.4              | 1,175.1           | 0.03     |
| <b>Robust LP (<math>\lambda_3=1</math>)</b> | 212.7 | 29.0   | 144.8              | <b>671.7</b>      | 0.04     |

Once the budget binds, every budget-exhausting policy **ties on coverage** ( $E[u]$ ,  $u^{\text{worst}}$ ). The robust LP wins on **routing** — same surge,  $4.6\times$  less transfer. The deterministic LP under-spends (51 beds) and pays  $\sim 3\times$  more unmet.

# Results — Allocation Structure & Budget Frontier



- (a) Robust LP finds a compromise (non-trivial North West + London surge); proportional spreads mechanically, greedy concentrates
- (b) The cost-shortage frontier is the decision-relevant trade-off a planner navigates as the budget grows

## Sensitivity — Which Control Matters?

---

- **Budget  $B$  dominates.**  $0.20 \rightarrow 0.10$  **doubles**  $E[u]$  and **triples**  $u^{\text{worst}}$ ;  $0.20 \rightarrow 0.30$  cuts  $E[u]$  to  $\sim \frac{1}{4}$ .
- **Tail weight  $\lambda_3$  is non-binding at  $B=0.20$**  — the expectation-minimiser already clears the high tail; only  $B \leq 15\%$  re-engages it (a diagnostic, not a defect).
- **Travel cap  $\tau=120$  min** cuts transfer burden 34% for a 3% rise in  $E[u]$ ;  $\tau \geq 240$  is slack.

**Across all 32 origins:** robust LP mean transfer **76 bed·km** vs **406 / 337** for proportional baselines, at tied mean unmet (2.9) — the routing gain holds across the whole demand distribution, not just the peak.

# Conclusions & Limitations

---

## Takeaways

- Matching the **forecaster loss to the operational cost of under-prediction** is what makes it useful
- Physics-informed parameter features beat lag-only learners on the **short-to-medium horizons** that drive surge decisions
- Once the budget binds, **routing efficiency** — not coverage — is the competitive dimension; **budget is the dominant lever**

## Limitations & next steps

- Detour-scaled travel times; well-mixed regional populations; a single realised allocation origin
- Co-monotone scenarios are deliberately over-conservative
- **Next:** trust-level instances (100s of sites → metaheuristics regain purpose), age-stratified pathways, all-origin allocation distributions

# Thank you — Questions?

Michael Ajao-Olarinoye · [olarinoyem@coventry.ac.uk](mailto:olarinoyem@coventry.ac.uk)

Centre for Computational Sciences and Mathematical Modelling, Coventry University

*A reproducible template for tying cost-asymmetric epidemic forecasting to transparent, robust capacity planning on open NHS data.*

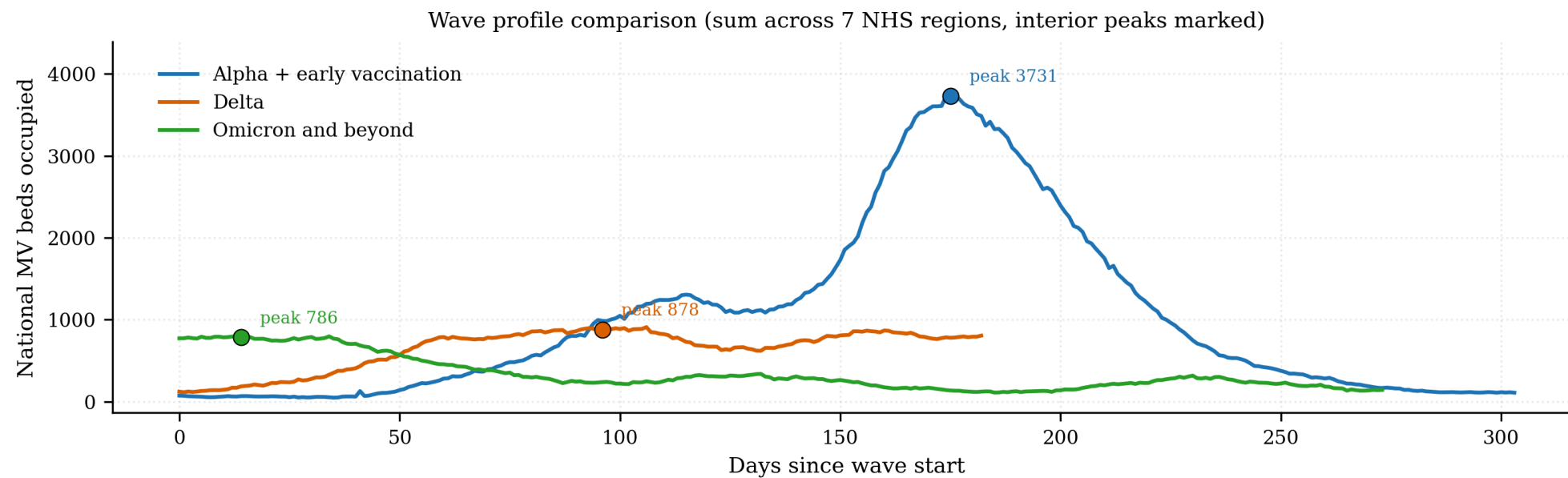
Code, configs & cleaned dataset — project repository.

## Backup — Key References

---

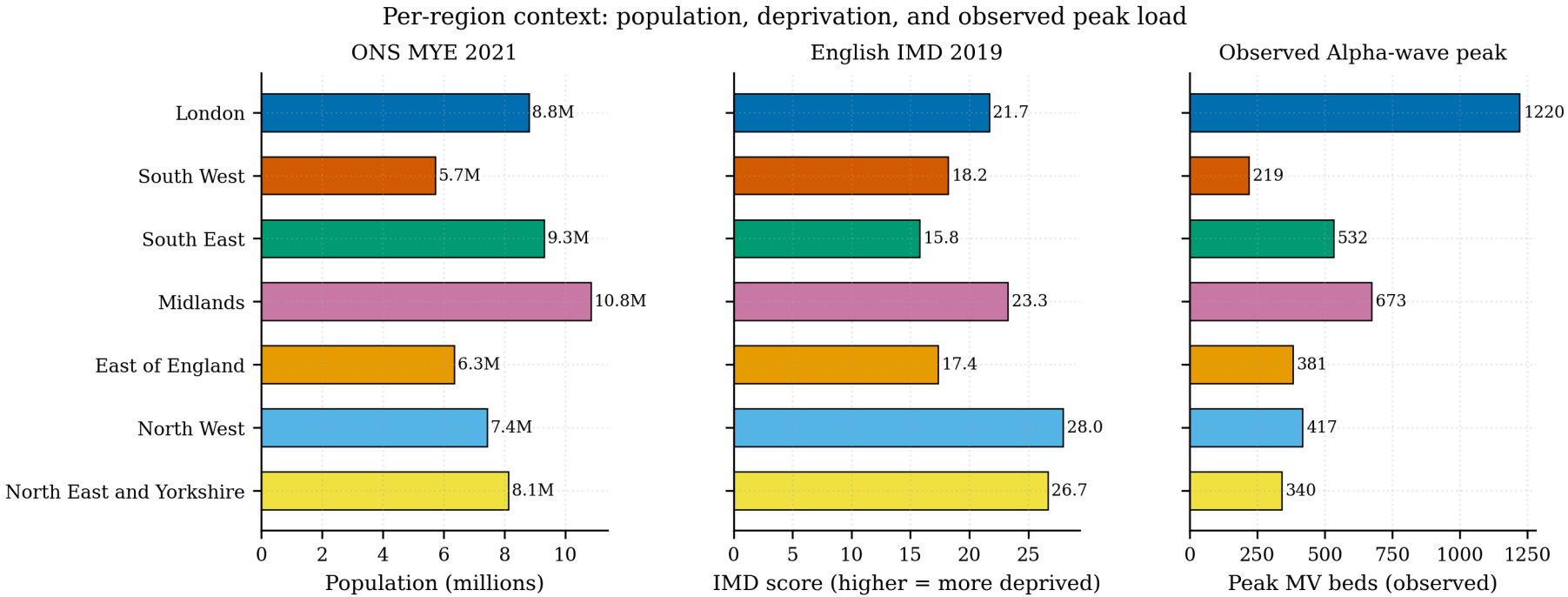
- Raissi et al. (2019) Physics-Informed Neural Networks — *J. Comput. Phys.* 378:686–707
- Koenker & Bassett (1978) Regression Quantiles — *Econometrica* 46(1):33–50
- Bracher et al. (2021) Evaluating epidemic forecasts (WIS) — *PLoS Comput. Biol.* 17(8)
- Bertsimas et al. (2022) DELPHI → vaccine-site MIP — *Naval Res. Logistics*
- Luo & Stellato (2024) Neural-ODE facility location (McCormick relaxation)
- Ajao et al. (2025) Hybrid physics-informed SEIRD forecasting — CRC Press

## Backup — Wave Overlay (Regime Shift)



Alpha / Delta / Omicron differ markedly in case-to-hospitalisation and H-to-C ratios — motivating the held-out Omicron test split.

# Backup — Regional Context



Geography of the seven NHS England regions underpins the inter-centroid travel-time parameterisation and the mutual-aid link structure.